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Body Measurement Accuracy and Pose Detection

Research Report

Introduction

Virtual Try-On systems are transforming the online shopping experience by allowing users to visualize how clothes, shoes, jewellery, and accessories would look on them before making a purchase. A key component of such systems is the ability to accurately understand the human body from images or videos. This is achieved through pose detection and body measurement estimation technologies.

Pose detection is a computer vision technique that identifies key points or landmarks on the human body. These landmarks represent different body parts such as the shoulders, elbows, hips, knees, and ankles. By analyzing these landmarks, it becomes possible to estimate body measurements and generate personalized recommendations for clothing sizes. Accurate body measurement estimation improves fitting accuracy, reduces product returns, and enhances customer satisfaction.

This report explores several popular pose detection models and evaluates their suitability for virtual try-on applications.

Pose Detection Technology

Pose detection refers to the process of identifying and tracking specific points on the human body using artificial intelligence and computer vision algorithms. Modern pose detection systems can analyze images and videos in real time and generate a skeletal representation of the human body.

In virtual try-on systems, pose detection helps identify body structure, posture, and proportions. These landmarks can then be used to estimate body measurements such as height, shoulder width, chest size, waist size, and hip size. Accurate pose estimation is essential for creating realistic virtual fitting experiences.

The effectiveness of pose detection directly influences the quality of avatar generation, clothing alignment, and garment fitting within AR and VR environments.

MediaPipe Pose

MediaPipe Pose is a pose estimation framework developed by Google. It is designed for real-time body tracking and can detect more than thirty body landmarks with high speed and accuracy. The framework is optimized for mobile devices and can operate efficiently even on smartphones with limited computational resources.

One of the primary advantages of MediaPipe Pose is its ability to perform real-time tracking while maintaining low latency. This makes it highly suitable for applications such as virtual try-on systems, fitness tracking, and augmented reality experiences. The model is lightweight and can be integrated easily into mobile and web applications.

Although MediaPipe Pose performs well under normal conditions, its accuracy may decrease when body parts are occluded or when the user is positioned at unusual angles. Nevertheless, it remains one of the most practical solutions for commercial applications.

OpenPose

OpenPose is a widely recognized pose estimation framework developed by Carnegie Mellon University. It is capable of detecting multiple people simultaneously and provides highly accurate landmark detection. OpenPose has become a standard tool in academic research and advanced computer vision applications.

The framework is known for its precision and detailed body tracking capabilities. It can accurately identify body joints, hand positions, and facial landmarks, making it suitable for applications requiring high-quality human pose estimation.

Despite its excellent accuracy, OpenPose requires significant computational resources and is not ideal for real-time mobile applications. Its processing requirements make it more suitable for research environments and high-performance computing systems.

MoveNet

MoveNet is a pose detection model developed by Google that focuses on speed and efficiency. The model is specifically optimized for mobile devices and real-time applications. It provides accurate body landmark detection while maintaining extremely fast processing speeds.

MoveNet is particularly useful in situations where low latency is required. The model performs efficiently on smartphones and embedded devices, making it a strong candidate for mobile virtual try-on systems.

Although MoveNet prioritizes speed, it may provide fewer landmark details compared to more complex frameworks such as OpenPose. However, its excellent performance-to-resource ratio makes it highly attractive for consumer-facing applications.

BlazePose

BlazePose is another pose estimation framework developed by Google. It offers a balance between speed and accuracy and has been widely adopted in fitness, health, and augmented reality applications. The framework is capable of tracking full-body movements in real time and performs well across various environments.

BlazePose provides detailed body landmark detection while maintaining efficient performance. Its ability to track body posture accurately makes it valuable for applications involving human movement analysis and virtual fitting systems.

The framework performs best when provided with clear images and sufficient lighting conditions. Poor image quality may affect the accuracy of landmark detection. Nevertheless, BlazePose remains one of the strongest options for real-time body tracking.

Body Measurement Estimation

Body measurement estimation involves calculating physical body dimensions using detected pose landmarks. Once body joints and key points are identified, mathematical relationships between these points can be used to estimate measurements such as height, shoulder width, chest circumference, waist circumference, hip circumference, arm length, and leg length.

In virtual try-on systems, these measurements play a critical role in recommending appropriate clothing sizes and improving garment fitting accuracy. Measurement estimation also supports the creation of realistic 3D avatars that closely resemble the user's body structure.

The accuracy of body measurement estimation depends heavily on the quality of pose detection, image resolution, camera positioning, and lighting conditions. Advanced algorithms can further improve estimation accuracy by combining pose data with machine learning models trained on body measurement datasets.

Challenges in Body Measurement Accuracy

Although pose detection technology has advanced significantly, several challenges continue to affect measurement accuracy. Clothing can hide important body contours, making it difficult to estimate true body dimensions. Loose garments often create additional uncertainty in measurement calculations.

Camera angle also plays an important role. Images captured from improper viewpoints may distort body proportions and reduce measurement reliability. Similarly, poor lighting conditions can affect landmark visibility and lead to inaccurate body tracking.

Occlusions represent another major challenge. When body parts are partially hidden by objects or other people, pose detection models may struggle to identify key landmarks correctly. Variations in posture and body positioning can further influence measurement outcomes.

To address these challenges, many modern systems combine pose estimation with depth sensing, multiple camera views, and machine learning techniques to improve overall accuracy.

Comparative Analysis of Pose Detection Models

The four pose detection frameworks examined in this report each offer distinct advantages. OpenPose provides the highest level of accuracy and detailed landmark detection but requires substantial computational resources. MediaPipe Pose offers an excellent balance between accuracy and performance, making it highly suitable for commercial applications. MoveNet prioritizes speed and mobile efficiency, making it ideal for smartphone-based solutions. BlazePose provides strong tracking performance and reliable body landmark estimation while maintaining real-time capabilities.

For large-scale virtual try-on platforms, selecting the appropriate model depends on project requirements, target devices, and performance expectations.

Recommendation for Raritone

After evaluating the available pose detection technologies, MediaPipe Pose and BlazePose emerge as the most suitable options for the Raritone Virtual Try-On platform. Both frameworks provide accurate body landmark detection, real-time performance, and compatibility with mobile devices.

MediaPipe Pose is particularly attractive due to its lightweight architecture, ease of integration, and strong industry adoption. BlazePose offers comparable accuracy while maintaining efficient performance for augmented reality applications.

These frameworks can support body measurement estimation, avatar generation, and virtual garment fitting while ensuring smooth user experiences across a wide range of devices.

Conclusion

Pose detection and body measurement estimation are fundamental technologies in modern virtual try-on systems. They enable accurate understanding of human body structure, allowing personalized fitting experiences and improved clothing recommendations. MediaPipe Pose, OpenPose, MoveNet, and BlazePose each contribute unique strengths to this field.

Among the evaluated solutions, MediaPipe Pose and BlazePose provide the most practical balance between accuracy, speed, and scalability for real-world deployment. Their ability to perform real-time body tracking on mobile devices makes them ideal choices for the Raritone Virtual Try-On platform. By leveraging these technologies, Raritone can enhance user engagement, improve fitting accuracy, and deliver a more immersive online shopping experience.